

SHAORYANG, China

WMO: 577660

Lat: 27.1850N

Lon: 111.4535E

Elev: 1018

StdP: 14.16

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.8	32.0	16.6	13.2	41.7	21.1	16.5	41.4	12.6	44.2	11.1	44.6	3.5	20	0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	95.5	77.5	93.5	77.5	91.6	77.3	80.4	88.8	79.6	87.9	78.9	87.1	5.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.5	153.7	83.2	77.6	148.9	83.0	76.8	145.2	82.8	45.1	88.9	44.2	88.2	43.4	87.3	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.2	10.2	8.8	DB	26.7	98.4	3.1	2.1	24.5	99.9	22.7	101.1	20.9	102.2	18.7	103.8	(4)
			WB	25.5	81.7	2.9	1.2	23.4	82.6	21.7	83.3	20.1	83.9	17.9	84.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.3	42.5	46.5	54.3	64.8	72.5	78.2	83.5	82.6	76.2	66.8	56.6	46.4
	DBStd	15.46	6.68	8.29	7.99	7.67	5.60	4.70	4.17	4.54	6.11	6.21	7.01	5.66
	HDD50	610	250	154	45	2	0	0	0	0	0	0	20	139
	HDD65	2561	699	520	343	97	6	0	0	0	3	53	263	577
	CDD50	5842	16	56	178	447	699	846	1039	1011	785	520	218	27
	CDD65	2318	0	2	11	91	240	396	574	546	338	108	12	0
	CDH74	22367	1	14	77	596	1683	3495	6891	6097	2888	561	64	0
	CDH80	8971	0	1	15	154	514	1207	3196	2675	1081	119	9	0
Wind	WSAvg	3.7	3.4	3.6	3.6	3.8	3.6	3.5	4.0	4.1	4.0	3.6	3.3	3.3
Precipitation	PrecAvg	51.3	2.2	3.2	4.2	7.3	8.0	7.8	4.1	4.2	2.5	3.5	2.9	1.5
	PrecMax	71.0	5.1	6.1	7.8	14.8	18.5	15.2	11.1	9.5	9.2	8.2	7.4	3.3
	PrecMin	36.3	0.2	0.4	1.0	3.4	2.6	2.0	0.7	0.2	0.0	0.0	0.0	0.0
	PrecStd	8.6	1.2	1.5	1.5	2.5	3.7	3.4	2.5	2.6	2.1	2.5	2.0	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.0	76.0	81.6	88.8	91.4	93.3	97.6	98.4	95.2	87.8	80.6	67.3
		MCWB	57.6	62.3	68.4	71.2	74.1	78.3	78.2	77.4	75.9	69.8	66.6	54.1
	2%	DB	61.2	69.1	75.2	84.5	88.3	91.1	95.6	95.8	91.9	83.2	75.2	63.0
		MCWB	51.4	58.8	63.4	70.5	73.0	77.5	78.0	77.3	76.0	68.7	62.1	51.8
	5%	DB	56.4	63.4	70.8	80.8	85.6	88.9	93.8	93.4	89.1	79.8	71.3	59.3
		MCWB	48.5	55.1	61.1	69.0	72.6	77.1	77.8	77.6	75.0	66.4	60.0	49.7
	10%	DB	52.4	58.3	66.5	77.0	82.6	86.7	91.9	91.0	86.3	76.8	67.7	55.5
		MCWB	46.4	51.9	58.5	67.3	71.8	76.3	77.7	77.5	73.9	65.1	58.0	48.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.5	63.2	70.1	74.8	78.3	80.3	81.6	81.5	79.7	75.3	68.8	58.6
		MCDB	65.9	73.7	79.8	84.5	86.1	87.8	89.9	89.6	87.9	83.2	76.7	62.3
	2%	WB	53.0	59.9	66.0	72.7	76.7	79.2	80.5	80.3	78.1	71.9	65.2	54.3
		MCDB	58.2	68.1	72.7	81.3	83.4	87.0	89.2	88.7	86.7	77.5	71.6	59.8
	5%	WB	49.6	56.4	62.9	71.0	75.3	78.5	79.7	79.4	77.1	69.9	62.6	51.3
		MCDB	54.9	61.9	68.4	78.0	81.8	86.1	88.3	87.6	85.4	76.1	68.1	57.0
	10%	WB	46.8	52.9	59.9	69.0	73.7	77.7	79.0	78.6	75.8	68.0	60.0	49.2
		MCDB	51.7	57.4	64.8	74.8	79.5	84.5	87.4	86.6	84.0	74.1	65.3	54.3
Mean Daily Temperature Range	5% DB	MDBR	9.3	10.3	11.5	12.8	12.6	11.2	12.7	12.8	13.0	12.4	12.3	11.1
		MCDBR	19.0	20.8	20.3	19.9	18.2	14.7	16.1	16.6	17.3	18.1	19.6	20.6
		MCWBR	10.7	11.3	9.7	8.0	6.0	4.5	3.7	3.6	4.7	6.0	8.2	10.4
	5% WB	MCDBR	16.1	18.4	17.6	16.9	13.7	12.6	14.1	13.9	15.0	13.4	15.3	15.4
MCWBR		9.7	11.2	10.2	8.9	6.5	4.8	4.4	4.2	4.7	6.1	7.7	8.7	
Clear-Sky Solar Irradiance	taub	0.682	0.714	0.773	0.747	0.662	0.643	0.569	0.654	0.686	0.741	0.674	0.662	
	taud	1.497	1.476	1.420	1.509	1.703	1.759	1.957	1.724	1.630	1.478	1.564	1.549	
	Ebn at Noon	177	187	187	199	218	220	237	217	203	178	178	174	
	Edh at Noon	82	89	100	93	77	72	59	74	80	88	75	75	
All-Sky Solar Radiation	RadAvg	515	615	817	1034	1176	1241	1625	1424	1164	921	750	601	
	RadStd	112	167	105	142	188	176	180	112	124	134	123	141	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)		+0.54	N/A	N/A	+1.06	N/A	N/A	N/A	N/A	+245	+133

Nomenclature: See separate page